ST5201X: Statistical Foundations of Data Science
(Semester 1, 2026/2027)

- **Lecturer:** Zhang, Yao
Department of Statistics & Data Science
Room: L7-106, Faculty of Science
Email: yaozhang@nus.edu.sg
- **Time and Venue:** 7:00 - 10:00 PM (Thursday (L1), Friday (L2)), Lecture Theatre 34 (LT34), Tutorial starts from Week 3 during regular lecture time.
- **Textbooks:**
  1. *Mathematical statistics and data analysis* by John Rice. Third edition. Brooks/Cole, Cengage Learning.
- **Assessment:** Homework: 50%; Final exam: 50%
- **Homework policy:** Submission on Canvas via a PDF file. **DO NOT** email your homework to me. Solutions will be posted on Canvas. **No late homework will be accepted.**
- Exam policy: Final exam date: **TBA**. Final exam will be CLOSED BOOK. However, each student is allowed to prepare **ONE** A4 sheet of notes (two-sided). One calculator without programming function is **allowed**. Computers are **prohibited**. Past exam papers are not available. **Failing to take the final exam means failing the course.**

Course Schedule		
Date	**Topic**	**Remarks**
Week 1 (Aug 13 & 14)	Introduction & Probability	
Week 2 (Aug 20 & 21)	Random Variables	
Week 3 (Aug 27 & 28)	Random Variables & Joint Distributions	Homework 1 Due
Week 4 (Sep 3 & 4)	Joint Distributions & Expected Values	
Week 5 (Sep 10 & 11)	Expected Values	
Week 6 (Sep 17 & 18)	Limit Theorems	Homework 2 Due
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Week 7 (Oct 1 & 2)	Estimation of Parameters and Fitting of Probability Distributions	
Week 8 (Oct 8 & 9)	Estimation of Parameters and Fitting of Probability Distributions & Testing Hypotheses	
Week 9 (Oct 15 & 16)	Testing Hypotheses	Homework 3 Due
Week 10 (Oct 22 & 23)	Distributions Derived from the Normal Distribution & Predictive Analytics	
Week 11 (Oct 29 & 30)	Nonparametric estimation	Homework 4 Due
Week 12 (Nov 5 & 6)	Nonparametric estimation & review	
Week 13 (Nov 12 & 13)	**TBA**	Homework 5 Due